

Answer Key & Teacher Notes

Session 1 – Fractions, Decimals, Percents • Session 2 – Compare and Order

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

Fractions, Decimals, Percents

TALK PROMPTS

- How does writing an equivalent fraction with a denominator of 100 help you read the percent?
- Why is 0.5 equal to 50% and not 5%? Use the word hundredths.

PARTNER TASK

Free Throws and Phone Battery. One partner writes the fraction or decimal; the other writes the percent and explains the scaling. Switch roles with Homework Check: 26 of 40 problems ($26/40 = 13/20 = 0.65 = 65\%$).

WATCH FOR

- Reading 0.08 as 80% or 0.7 as 7% (misplacing the tenths and hundredths digits).
- Writing the part as the percent without scaling to 100 (26 of 40 called “26%”).
- Using the wrong whole: comparing 8 red to 12 blue beads instead of 8 to the total of 20.

SESSION 2 • TEACHER NOTES

Compare and Order

TALK PROMPTS

- Which form did your group choose for comparing, and what made that form easier to use?
- How can you tell $4/5$ lands between 0.7 and 86% without dividing?

PARTNER TASK

Quiz Scores. One partner writes 0.76, 18/25, and 79% as percents (76%, 72%, 79%); the other orders them least to greatest and checks. Switch roles with Rainfall: 0.12, 3/20, and 9%.

WATCH FOR

- Comparing digits instead of values ($0.4 < 39\%$ “because $4 < 39$ ”).
- Treating a longer decimal as larger, or reading 0.6 as 6%.
- Missing ties: 0.6 and $3/5$ are equal and should sit side by side.

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.RP.A.3c	1	2	1	1	2	2	1	3	3	1	1
S1 • Form B	6.RP.A.3c	1	2	1	1	2	2	1	3	3	1	1
S2 • Form A	6.RP.A.3c	1	2	1	1	2	2	2	3	3	1	1
S2 • Form B	6.RP.A.3c	1	2	1	1	2	2	2	3	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

percent: A ratio that compares a number to **100**.

equivalent: Names the **same** amount in a different way.

hundredths: Parts out of 100; the **second** place after the decimal point.

- Fraction to percent: make an equivalent fraction with a denominator of **100**.
- Decimal to percent: read the decimal in **hundredths**, so $0.4 = 0.40 = 40\%$.
- Percent to fraction: write the percent over 100, then **simplify**.

Example: Fraction, part over whole: **7/20**. $20 \times 5 = 100$, so $7/20 = 35/100$. $35/100 = \mathbf{0.35}$ as a decimal. Answer: $7/20 = 0.35 = \mathbf{35\%}$

YOU TRY

1. a. 60% b. 8% c. 70%
2. 0.45 and $45/100 = 9/20$

EXIT TICKET

1. a. 9% b. 65%
2. $18/25 = 72/100 = 0.72 = 72\%$
3. **C** ($4/100$)

PRACTICE FORM A

1. a. 60% b. 45% c. 8% d. 12% e. 72% f. 25%
2. Hamster: $16/50 = 8/25 = 0.32 = 32\%$. Fish: $24/50 = 12/25 = 0.48 = 48\%$. Turtle: $50 - 40 = 10$ votes, $10/50 = 1/5 = 0.2 = 20\%$.
3. a. $12/16 = 3/4$ b. 75%
4. a. 60% b. 40%
5. $26/40 = 13/20 = 65/100 = 0.65 = 65\%$
6. $8/20 = 40/100 = 40\%$, so 0.4
7. a. 45% b. 0.45 c. $45/100 = 9/20$
8. a. A 45%, B 44%, C 46% b. B, A, C
9. The 8 is in the hundredths place, so $0.08 = 8/100 = 8\%$. 80% would be 0.80 (8 tenths).
10. **C** (80%)
11. **B** (0.25)

Self-check word **PERCENT**

PRACTICE FORM B

1. a. 90% b. 55% c. 4% d. 24% e. 38% f. 75%
2. Museum: $14/40 = 7/20 = 0.35 = 35\%$. Zoo: $18/40 = 9/20 = 0.45 = 45\%$. Nature center: $40 - 32 = 8$ votes, $8/40 = 1/5 = 0.2 = 20\%$.
3. a. $14/20 = 7/10$ b. 70%
4. a. 35% b. 65%
5. $22/40 = 11/20 = 55/100 = 0.55 = 55\%$
6. $6/20 = 30/100 = 30\%$, so 0.3
7. a. 64% b. 0.64 c. $64/100 = 16/25$
8. a. A 62%, B 60%, C 61% b. B, C, A
9. The 7 is in the tenths place: $0.7 = 0.70 = 70/100 = 70\%$. 7% would be 0.07.
10. **C** (30%)
11. **B** (0.75)

Self-check word **DECIMAL**

GUIDED NOTES

common form: Every value written the **same** way before you compare.

benchmark: An easy number you know well, used to **compare** other numbers.

order: To list numbers from **least** to greatest, or greatest to least.

- To compare a fraction, a decimal and a percent, first write them all as **percents**.
- After you order the percents, write the list with the **original** numbers.

Example: $0.7 = 0.70 = 70\%$ $3/5 = 60/100 = 60\%$

Compare: $60\% < 65\% < 70\%$ Answer: Least to greatest:

3/5, 65%, 0.7

YOU TRY

- a. $>$ b. $=$ c. $<$
- 62%, 60%, 61%, so $3/5$, 61%, 0.62

EXIT TICKET

- a. $>$ b. $<$ c. $=$
- 35%, 37.5%, 36%, so 0.35, 36%, $3/8$
- B** ($3/4$)

PRACTICE FORM A

- a. $>$ b. $<$ c. $=$ d. $<$ e. $<$ f. $>$
- a. Red 90%, Blue 80%, Green 86%, Gold 70% b. Gold, Blue, Green, Red
- a. 76%, 72%, 79% b. $18/25$
- a. 12%, 15%, 9% b. $3/20$
- No. $0.4 = 0.40 = 40\%$, and $40\% > 39\%$, so 0.4 is greater.
- Heroes 35%, villains 40%, sidekicks 25% (total 100%); villains are the largest group.
- 36%, 36%, 34%. Order: 34%, then 0.36 and $9/25$ tie at 36%.
- a. Ana 62%, Ben 62.5%, Cal 61%, Dee 60% b. Dee, Cal, Ana, Ben
- Owen compared digits, not values. $0.6 = 60\%$, $3/5 = 60/100 = 60\%$, so 0.6 and $3/5$ tie. Order: 0.6 and $3/5$ (tie at 60%), then 61%.
- A** (0.6)
- C** ($1/4$)

Self-check word VALUES

PRACTICE FORM B

- a. $>$ b. $<$ c. $=$ d. $<$ e. $<$ f. $>$
- a. Maple 85%, Oak 90%, Pine 83%, Cedar 75% b. Cedar, Pine, Maple, Oak
- a. 88%, 84%, 86% b. $21/25$
- a. 14%, 16%, 11% b. $4/25$
- No. $0.3 = 0.30 = 30\%$, and $30\% > 29\%$, so 0.3 is greater.
- Animals 45%, stars 30%, hearts 25% (total 100%); animals are the largest group.
- 48%, 48%, 46%. Order: 46%, then 0.48 and $12/25$ tie at 48%.
- a. Eli 84%, Fay 87.5%, Gus 86%, Hana 80% b. Hana, Eli, Gus, Fay
- $0.8 = 80\%$ and $4/5 = 80/100 = 80\%$, so they are equal, not less than. Correct: $0.8 = 4/5 < 83\%$ (0.8 and $4/5$ tie at 80%).
- A** (0.7)
- C** ($3/8$)

Self-check word SHARES

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 · FRACTIONS, DECIMALS, PERCENTS

1. a. $35/100 = 7/20$ b. 0.35 c. 35%
2. $1/4 = 0.25 = 25\%$; $7/10 = 0.7 = 70\%$; $2/25 = 0.08 = 8\%$; $13/20 = 0.65 = 65\%$; $3/5 = 0.6 = 60\%$
3. $11/25 = 44/100 = 0.44 = 44\%$
4. She just wrote the digits 3 and 4 side by side; that is not the same value. $3/4 = 75/100 = 0.75 = 75\%$.

SHEET 2 · COMPARE AND ORDER

1. a. $>$ b. $<$ c. $<$
2. a. $7/10$, 0.72, 75% b. 0.2, 22%, $1/4$ c. $6/5$, 125%, 1.3
3. A = 55%, B = 60%, C = 52%. Order: Room C, Room A, Room B (Room B greatest).
4. $3/5 = 60\%$ and $0.7 = 70\%$, so any value between 60% and 70%. Sample: $13/20 = 0.65 = 65\%$.
5. Comparing single digits does not compare values; change to one form. $3/8 = 0.375 = 37.5\%$ and $0.4 = 40\%$, so $3/8 < 0.4$.
6. **C** (0.35, 38%, $2/5$)
7. **C** (64%)